

Business Analytics in Action: Which Survey Relationships Survive Adjustment?

pypasso: covariate-adjusted association between mixed-type outcomes, validated on six real surveys

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Key finding: respondent composition can explain half of a raw survey association

Across **six real surveys** ($n = 1,118$ to $107,736$), the unchanged **pypasso** pipeline placed continuous scores, 1–5 ratings, and 0–10 scales on **one residual scale**. Strong associations **survived** adjustment ($0.48 \rightarrow 0.35$; $-0.49 \rightarrow -0.38$), respondent composition explained **52% of one association**, and a near-zero association remained near zero.

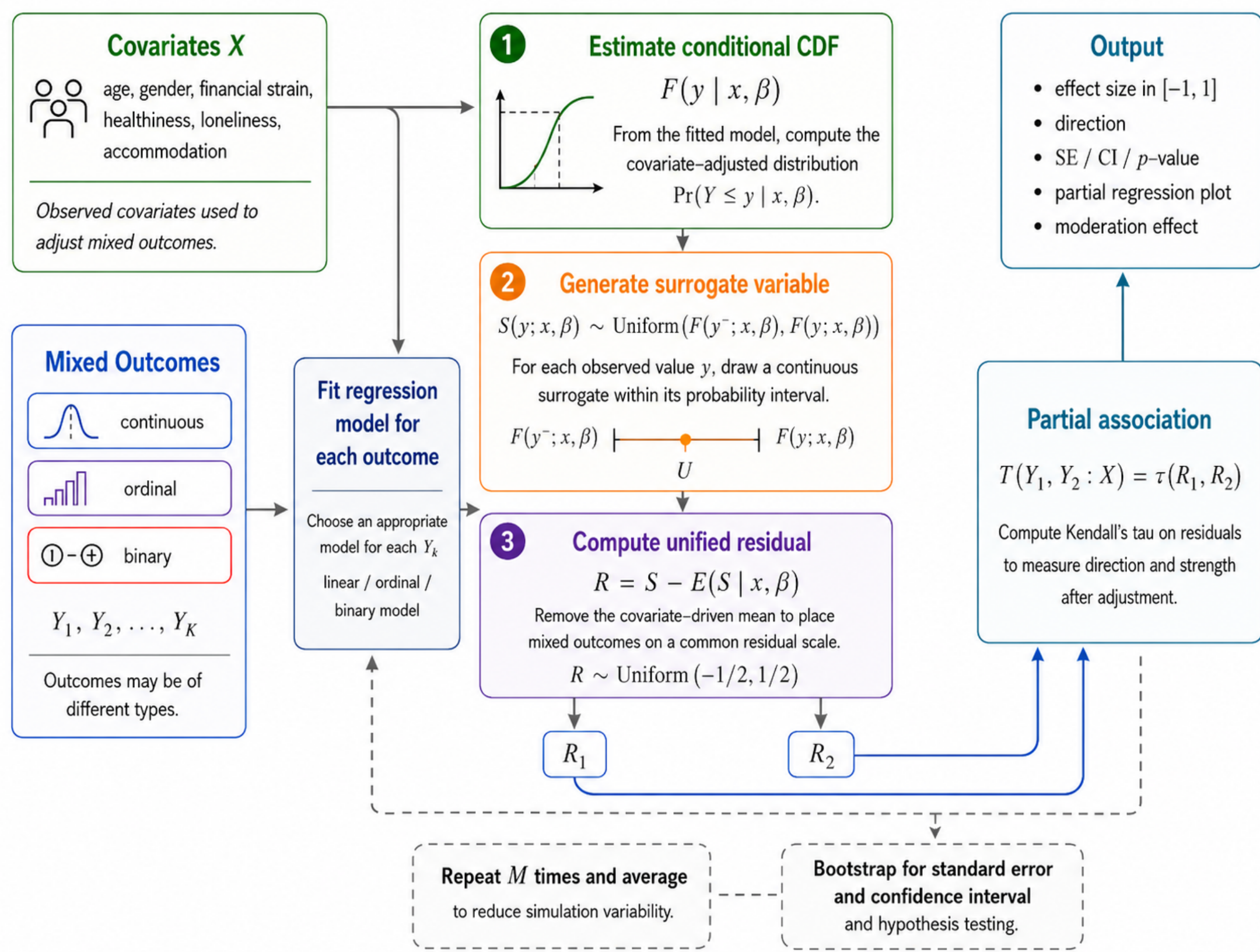
1. The Problem: Mixed Scales, Hidden Composition

“Sometimes the averages look calm, but the relationships *inside* the data reveal where institutions need to act.” Analysts often ask whether two outcomes move together, such as *satisfaction* and *renewal* or *well-being* and *anxiety*. Three issues make this difficult:

- Outcomes live on **different scales** (a continuous score, a 1–5 rating, or a yes/no flag), making standard effect sizes difficult to compare.
- Any apparent link may be **composition**: shared background factors (age, income, politics) can create (or hide) it.
- Regression-based moderation estimates are not stable here: *reversing* which outcome is the “response” shifted the estimate from -34.5% to -9.3% to -4.6% (Li et al., 2024).

Goal: port the rigorous fix, the surrogate-residual method of Li et al. (AOAS 2024), previously R-only (**P**Asso), into a documented, installable, real-data-tested Python package a new analyst can run unaided.

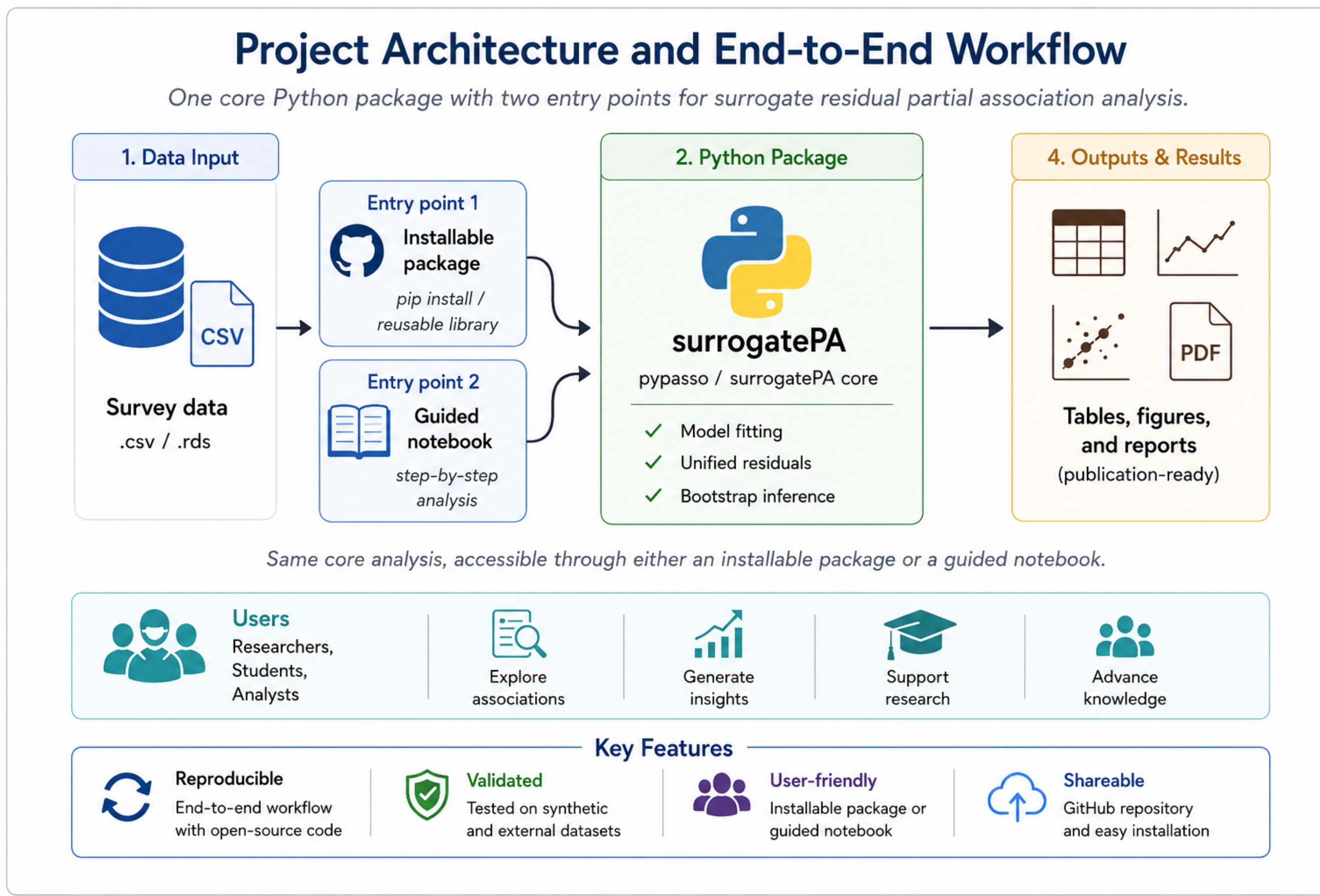
2. The Method: One Residual Scale for Every Outcome



- Fit** a model per outcome given covariates X (LM continuous, ACL ordinal).
- Unify**: draw a continuous surrogate in each observation’s fitted probability interval, subtract its covariate-driven mean $\Rightarrow$ residual $R \sim \text{Uniform}(-\frac{1}{2}, \frac{1}{2})$ on a common scale.
- Compare**: partial association $= \tau(R_1, R_2)$, symmetric in the two outcomes; M surrogate draws + bootstrap give SE, 95% CI, p -value.

marginal = raw link · **partial** = after adjusting for X · **reduction** = share explained by X

3. The Tool: One Core, Two Entry Points



One core, two doors: a tested, versioned library for **pip** users, or a guided, zero-setup notebook. Same data, same seed $\Rightarrow$ **identical numbers** on both routes.

2 37 6 108K 0
entry points passing tests real surveys max sample size code changes needed

4. Main Contribution: From Research Method to Reusable Python Tool

The contribution is more than a line-by-line code translation. The team delivered:

- Two usable entry points**: a versioned, installable Python package and a self-contained guided notebook.
- Real-data readiness**: preprocessing for missing-value sentinels, malformed CSVs, nonstandard delimiters, and legacy encodings.
- Reproducibility and reliability**: 37 automated tests, fixed seeds, 13 actionable error messages, and manual-style documentation.
- Evidence beyond the original example**: the unchanged workflow ran across six real surveys from 1.1K to 108K observations.

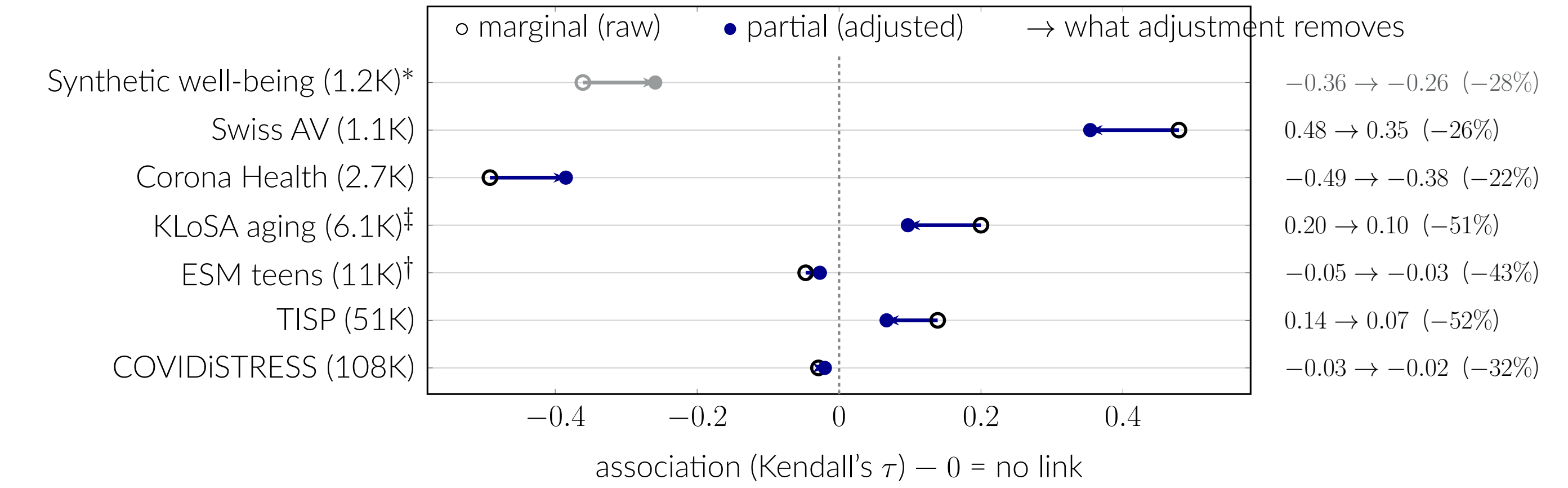
5. From Synthetic Baseline to Real Data (v0.3.0)

Development anchored on a **bundled synthetic replica** of the paper’s student well-being survey (well-being $\times$ anxiety, given health, loneliness, accommodation, financial strain): **run_example()** reproduces its reference numbers in one call ($-0.361 \rightarrow -0.259$), matching R’s **P**Asso within 0.005, the regression test every change must pass. External testing was then a **feedback loop**: surveys 1–2 exposed friction, v0.3.0 fixed it, and later surveys exercised those same features while revealing additional edge cases.

Real-world mess (survey)	One-line fix in pypasso
–99 “no answer” sentinels (Swiss AV)	<code>prepare_data(na_values=[–99], dropna=True)</code>
unquoted text column breaks ~1,700 rows (Corona Health)	<code>load_data(usecols=[...])</code>
European format ; , + BOM (TISP)	<code>load_data(sep=";", decimal=",")</code>
legacy Latin-1 encoding (COVIDiSTRESS)	<code>load_data(encoding="latin-1")</code>

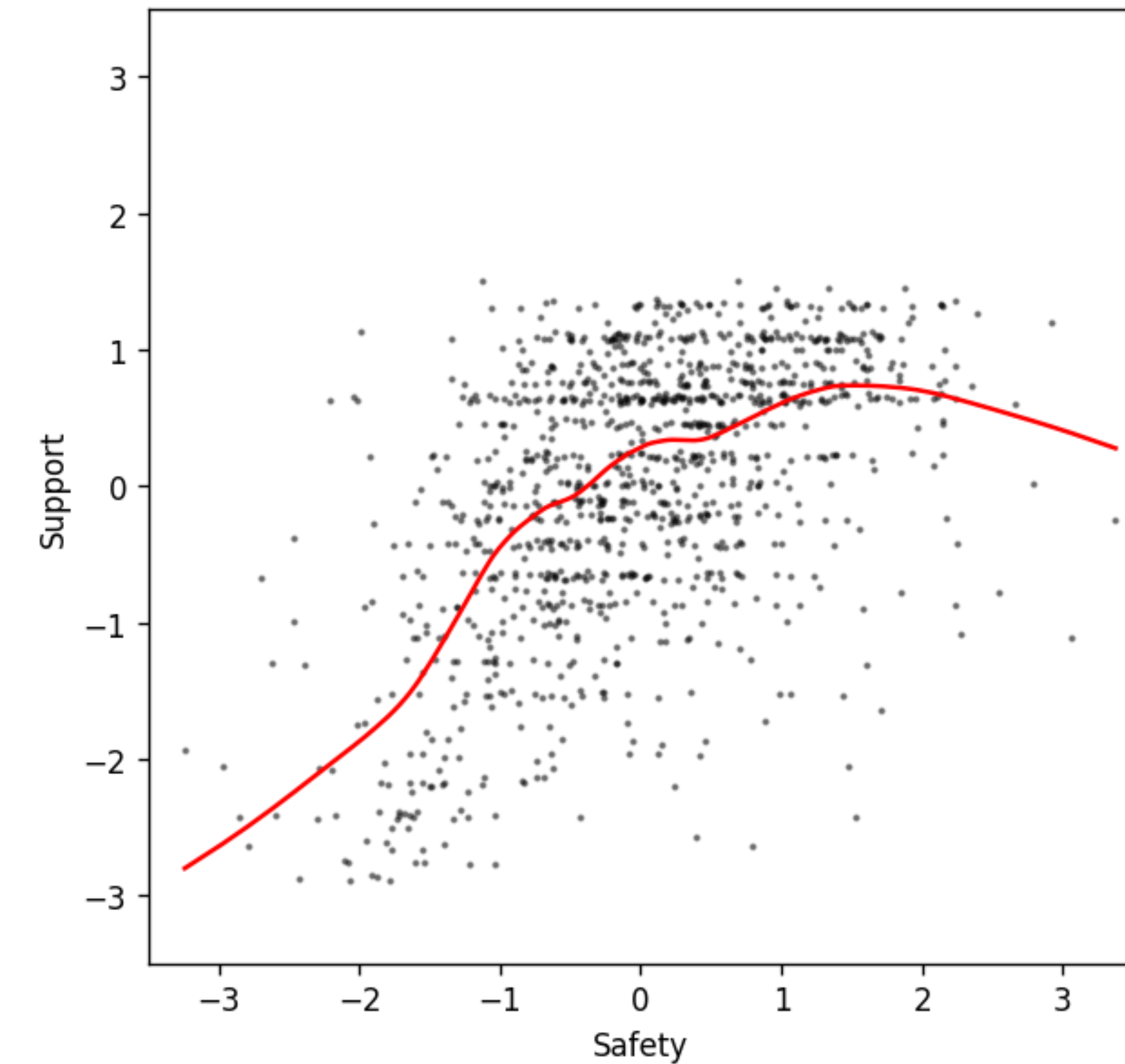
6. Validation: Six Surveys, One Unchanged Pipeline

After matching the synthetic baseline (gray row), the pipeline ran on six external surveys with **zero package source-code edits** and fixed random seeds:



Adjustment pulls every link toward zero; the key question is *how far*. Strong links survive; TISP and KLoSA are about half composition; the near-zero stays near-zero. All bootstrap 95% CIs exclude zero; **LM** $\leftrightarrow$ **ACL** swaps move **partial** by < 0.03 . *dev baseline; †,‡via James’s notebook.

7. Reading the Numbers



Swiss AV survey: each dot represents one respondent’s pair of unified residuals. The trend shows what remains *after* adjusting for age, gender, and education. Support for autonomous vehicles and perceived safety remain clearly associated (partial 0.35, 95% CI [0.31, 0.38]).

- Composition, made visible**: TISP’s trust–climate link is over half *political orientation*; KLoSA’s health–depression link is about half *age and gender*, the clearest demo of what “partial” means.
- COVIDiSTRESS**: at $n > 100K$ even -0.02 gets $p \approx 0$, so the honest summary is effect size, not significance; the package reports a null as a null.

Acknowledgments & Reference

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Reference. [1] Li, S., Fan, Z., Liu, I., Morrison, P. S., & Liu, D. (2024), “Surrogate method for partial association between mixed data with application to well-being survey analysis,” *The Annals of Applied Statistics*, 18(3), 2254–2276.